Dual-Weighted-Residual Adaptive Loop for Linear Time-Dependent PDEs

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1 Scope and Notation

The framework covers well-posed linear 1st-order-in-time equations that can be written in a Hilbert space as

$$M\partial_t u + \mathcal{L}(t)u = f, \qquad t \in (0, T], \tag{1}$$

with an initial condition and suitable boundary conditions.

Let $m : H \times H \to \mathbb{R}$ be a continuous symmetric positive-definite mass form and let $a(t;\cdot,\cdot) : V \times V \to \mathbb{R}$ be a measurable continuous bilinear form satisfying the assumptions required for well-posedness. Define

$$\begin{aligned}
\mathcal{X} &:= \{v \in L^2(0,T;V) : \partial_t v \in L^2(0,T;V^*)\},\\
\mathcal{Y} &:= \mathcal{X}.
\end{aligned}$$

2 Continuous Primal and Adjoint Problems

For $v \in \mathcal{Y}$, set

$$\mathcal{A}(u)(v) := \int_0^T [m(\partial_t u, v) + a(t; u, v)]\,\mathrm{d}t + m(u(0), v(0)), \tag{2}$$

$$\mathcal{F}(v) := \int_0^T \ell(t; v)\,\mathrm{d}t + m(u_0, v(0)). \tag{3}$$

The continuous primal problem is

$$\boxed{\text{Find } u \in \mathcal{X} \text{ such that } \mathcal{A}(u)(v) = \mathcal{F}(v) \quad \forall v \in \mathcal{Y}.} \tag{4}$$

The initial condition is encoded weakly by the last terms in equations (2) and (3).

Let the quantity of interest be $\mathcal{J}(u)$. The continuous adjoint is

$$\boxed{\text{Find } z \text{ such that } \mathcal{A}(w)(z) = \mathcal{J}(w) \quad \forall w \in \mathcal{X}.} \tag{5}$$

It is a terminal-value problem and is solved backward in time. This primal–adjoint construction and its goal-oriented identity are the foundation of the DWR method [1–3].

3 Time and Space Discretisation

3.1 Temporal Mesh and Jumps

Let

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad I_n = (t_{n-1}, t_n], \qquad k_n = t_n - t_{n-1}.$$

For a function that may be discontinuous at t_n, write

$$v_n^- := \lim_{t\uparrow t_n} v(t), \qquad v_n^+ := \lim_{t\downarrow t_n} v(t), \qquad [v]_n := v_n^+ - v_n^-.$$

The discontinuous Galerkin time space of degree r is

$$\mathcal{X}_k^r(V) := \{v : v|_{I_n} \in \mathbb{P}_r(I_n; V),\ 1 \le n \le N\}. \tag{6}$$

This jump convention and the dG time formulation below follow the classical analysis of dG time stepping for parabolic equations [4].

3.2 Spatial Meshes

Let $\mathcal{T}_h^n$ be a conforming spatial mesh on I_n, and let $V_h^n \subset V$ be its finite element space of degree p. Define

$$\mathcal{X}_{kh}^{r,p} := \{v : v|_{I_n} \in \mathbb{P}_r(I_n; V_h^n),\ 1 \le n \le N\}. \tag{7}$$

The broken dG bilinear form is

$$\begin{aligned}\mathcal{A}_{kh}(U)(v) := &\sum_{n=1}^{N} \int_{I_n} [m(\partial_t U, v) + a(t; U, v)]\,\mathrm{d}t \\ &+ \sum_{n=2}^{N} m([U]_{n-1}, v_{n-1}^+) + m(U_0^+, v_0^+),\end{aligned} \tag{8}$$

$$\mathcal{F}_{kh}(v) := \sum_{n=1}^{N} \int_{I_n} \ell(t; v)\,\mathrm{d}t + m(u_0, v_0^+). \tag{9}$$

The fully discrete primal solution is

$$\boxed{\mathcal{A}_{kh}(U_{kh})(v_{kh}) = \mathcal{F}_{kh}(v_{kh}) \quad \forall v_{kh} \in \mathcal{X}_{kh}^{r,p}.} \tag{10}$$

4 DWR Identity and Computable Enriched Weights

Extend $\mathcal{A}_{kh}$ consistently to exact and enriched functions and define

$$\rho_{kh}(U)(v) := \mathcal{F}_{kh}(v) - \mathcal{A}_{kh}(U)(v). \tag{11}$$

The discrete adjoint is the backward-in-time problem

$$\mathcal{A}_{kh}(w_{kh})(Z_{kh}) = \mathcal{J}(w_{kh}) \quad \forall w_{kh} \in \mathcal{X}_{kh}^{r,p}. \tag{12}$$

Proposition 1 (Linear DWR identity). *Assume that the primal problem, goal and discretisation are linear and consistent, and that quadrature and algebraic solves are exact. For every $\varphi_{kh} \in \mathcal{X}_{kh}^{r,p}$,*

$$\boxed{\mathcal{J}(u) - \mathcal{J}(U_{kh}) = \rho_{kh}(U_{kh})(z) = \rho_{kh}(U_{kh})(z - \varphi_{kh}).} \tag{13}$$

The first equality follows by testing the adjoint with $u - U_{kh}$; the second uses Galerkin orthogonality $\rho_{kh}(U_{kh})(\varphi_{kh}) = 0$. This is the linear DWR representation [1, 2]. For nonlinear operators or goals, the symmetric Lagrangian identity contains a dual residual, a factor 1/2, and a higher-order remainder [2]; those terms are absent from this linear specification.

Let Z^+ be a numerical adjoint in a strictly enriched space $\mathcal{X}_{kh}^+ \supsetneq \mathcal{X}_{kh}^{r,p}$. The direct enriched-weight estimator is

$$\eta_{\text{joint}} := \rho_{kh}(U_{kh})(Z^+ - Z_{kh}). \tag{14}$$

This replaces the exact defect $z - \varphi_{kh}$ by the numerical defect $Z^+ - Z_{kh}$. An enriched solve is more expensive but avoids relying solely on postprocessed interpolation. Reliability of enriched-space estimators is normally tied to a saturation assumption; interpolation recoveries need further recovery assumptions [5].

5 Separating Temporal and Spatial Goal Error

Let $U_k \in \mathcal{X}_k^r(V)$ be the time-semidiscrete solution: time is discretised but space is still V. Algebra gives

$$\boxed{\mathcal{J}(u) - \mathcal{J}(U_{kh}) = \underbrace{\mathcal{J}(u) - \mathcal{J}(U_k)}_{e_k \text{ (temporal)}} + \underbrace{\mathcal{J}(U_k) - \mathcal{J}(U_{kh})}_{e_h \text{ (spatial)}}.} \tag{15}$$

Separated temporal/spatial DWR representations based on this hierarchy were developed for parabolic problems and dynamic meshes by Schmich and Vexler [6], and related formulations are used for nonstationary flow and modern space–time DWR methods [7–9].

Let z_k be the time-semidiscrete adjoint and z_{kh} the fully discrete adjoint. In the linear case,

$$e_k = \rho_k(U_k)(z - z_k), \tag{16}$$

$$e_h = \rho_{kh}(U_{kh})(z_k - z_{kh}). \tag{17}$$

The residual need not be physically labelled temporal or spatial. The dual defect paired with it identifies the approximation-level error. Thus the volume residual

$$R = f - M\partial_t U_{kh} - \mathcal{L}U_{kh} \tag{18}$$

may occur in both separated estimators, once with a temporal dual defect and once with a spatial dual defect. Assigning its first term to time and its second to space would not be a DWR error split.

Construct the nested hierarchy

$$\mathcal{X}_{hk} \subset \mathcal{X}_{h^+k} \subset \mathcal{X}_{h^+k^+}, \tag{19}$$

where h^+ enriches only space and k^+ subsequently enriches time. Compute U_{hk}, Z_{hk}, the space-enriched pair U_{h^+k}, Z_{h^+k}, and the fully enriched adjoint $Z_{h^+k^+}$. Define

$$w_h := Z_{h^+k} - Z_{hk}, \qquad w_k := Z_{h^+k^+} - Z_{h^+k}, \tag{20}$$

and

$$\widehat{\eta}_h := \rho_{hk}(U_{hk})(w_h), \tag{21}$$

$$\widehat{\eta}_k := \rho_{h^+k}(U_{h^+k})(w_k), \tag{22}$$

$$\widehat{\eta}_{\text{split}} := \widehat{\eta}_k + \widehat{\eta}_h. \tag{23}$$

The hierarchy is motivated by equation (15) and the separated estimators of Schmich and Vexler [6]. However, equations (21) and (22) replace the unavailable space-exact semidiscrete objects by enriched numerical solves. They are not the identical theorem proved for the reconstruction operators of Schmich and Vexler [6]. Their reliability must be checked by enrichment and saturation tests [5].

6 Automated Localisation by Bubble Recovery and Directional Splitting

The bubble functions are used to convert the weak residual into computable polynomial representations of its cell, spatial-facet and temporal-interface parts.

$$\boxed{\rho_{kh}(U_{kh})(\,\cdot\,) \xrightarrow{\text{bubble/cone projection}} (R^{\text{vol}}, R^{\text{sf}}, R^{\text{tf}}) \xrightarrow{\text{weights } w_h, w_k} (\eta^h_{K,n}, \eta^k_n).} \tag{24}$$

Here sf and tf denote spatial-facet and temporal-facet residuals, respectively. This is the residual-recovery construction of Rognes and Logg [10], followed by a space–time split of the DWR weight [6].

For a general linear space–time variational problem, define

$$\rho_{kh}(U_{kh})(V) := \mathcal{F}_{kh}(V) - \mathcal{A}_{kh}(U_{kh}, V). \tag{25}$$

If Z is the exact adjoint and Z_{kh} is any discrete adjoint, the linear DWR identity is

$$\mathcal{J}(u) - \mathcal{J}(U_{kh}) = \rho_{kh}(U_{kh})(Z - Z_{kh}) \tag{26}$$

up to algebraic inconsistency and data-approximation terms [2]. In computation, $Z - Z_{kh}$ is replaced by a numerical enriched-adjoint difference; the bubbles are used to localise the residual acting on that difference.

6.1 Stationary Bubble and Cone Projection

The stationary construction is recalled first . Suppose that a local residual has the representation

$$\rho_K(v) = (R_K, v)_K + \sum_{F \subset \partial K} (R_{K,F}, v)_F. \tag{27}$$

Let K be a d-simplex with barycentric coordinates $\lambda_1^K, \ldots, \lambda_{d+1}^K$. Define the cell bubble

$$b_K := \prod_{i=1}^{d+1} \lambda_i^K, \qquad b_K|_{\partial K} = 0. \tag{28}$$

For a chosen polynomial recovery space Q_K^p, the recovered cell residual $R_K^B \in Q_K^p$ is defined by

$$\boxed{(R_K^B, b_K \phi)_K = \rho_K(b_K \phi) \qquad \forall \phi \in Q_K^p.} \tag{29}$$

To recover the residual on a facet $F \subset \partial K$, introduce the facet cone

$$\beta_{K,F} := \prod_{i \in \mathcal{I}_F} \lambda_i^K, \tag{30}$$

where $\mathcal{I}_F$ contains the vertices belonging to F. The cone is nonzero on F and vanishes on every other facet of K. After the cell residual has been recovered, find $R_{K,F}^B \in Q_F^q$ from

$$\boxed{(R_{K,F}^B, \beta_{K,F}\psi)_F = \rho_K(\beta_{K,F}\psi) - (R_K^B, \beta_{K,F}\psi)_K \qquad \forall \psi \in Q_F^q.} \tag{31}$$

Thus the cell part is recovered first and subtracted before the remaining facet distribution is projected.

6.2 Space–Time Extension of the Residual Recovery

Let $I_n = (t_{n-1}, t_n]$, $k_n = t_n - t_{n-1}$, and

$$Q_{K,n} := K \times I_n, \qquad \tau := \frac{t - t_{n-1}}{k_n}.$$

The weak residual on $Q_{K,n}$ may contain three geometrically distinct distributions:

$$\rho_{K,n}(V) = (R^{\mathrm{vol}}_{K,n}, V)_{Q_{K,n}} + \sum_{F \subset \partial K} (R^{\mathrm{sf}}_{K,F,n}, V)_{F \times I_n} + (R^{\mathrm{tf}}_{K,n-1}, V(t^+_{n-1}))_K. \tag{32}$$

For dG time discretisations, $R^{\mathrm{tf}}_{K,n-1}$ contains the one-sided state jump, the initial-condition mismatch when $n = 1$, and any mesh-transfer inconsistency.

The following tensor-product construction is the proposed space–time extension of the Rognes–Logg recovery.

Volume projection. Use the temporal interior bubble

$$b^t_n(\tau) := 4\tau(1 - \tau) \tag{33}$$

and the space–time cell bubble

$$B_{K,n}(x, t) := b_K(x) b^t_n(t).$$

It vanishes on every codimension-one face of $Q_{K,n}$. For a space–time polynomial recovery space $Q^{p_x,p_t}_{K,n}$, define $R^{B,\mathrm{vol}}_{K,n}$ by

$$\boxed{(R^{B,\mathrm{vol}}_{K,n}, B_{K,n}\Phi)_{Q_{K,n}} = \rho_{K,n}(B_{K,n}\Phi) \qquad \forall \Phi \in Q^{p_x,p_t}_{K,n}.} \tag{34}$$

All spatial-facet and temporal-interface terms vanish under this test, so equation (34) projects the complete volume density.

Spatial-facet projection. For $F \subset \partial K$, define the space–time facet cone

$$C^{\mathrm{sf}}_{K,F,n}(x, t) := \beta_{K,F}(x) b^t_n(t). \tag{35}$$

It is nonzero on $F \times I_n$, zero on the other spatial facets, and zero at both time endpoints. After the volume solve, recover $R^{B,\mathrm{sf}}_{K,F,n}$ from

$$\begin{aligned}&(R^{B,\mathrm{sf}}_{K,F,n}, C^{\mathrm{sf}}_{K,F,n}\Psi)_{F \times I_n} \\ &\quad = \rho_{K,n}(C^{\mathrm{sf}}_{K,F,n}\Psi) - (R^{B,\mathrm{vol}}_{K,n}, C^{\mathrm{sf}}_{K,F,n}\Psi)_{Q_{K,n}} \quad \forall \Psi \in Q^{q_x,q_t}_{F,n}.\end{aligned} \tag{36}$$

Temporal-interface projection. An interior temporal bubble cannot detect a jump because it vanishes at the endpoint. The temporal analogue of the facet cone is instead

$$C^{\mathrm{tf},-}_{K,n}(x, t) := b_K(x)(1 - \tau), \tag{37}$$

which is nonzero at t^+_{n-1}, zero at t^-_n, and zero on the spatial boundary. The recovered temporal-face residual is defined by

$$\begin{aligned}&(R^{B,\mathrm{tf}}_{K,n-1}, b_K\chi)_K \\ &\quad = \rho_{K,n}(C^{\mathrm{tf},-}_{K,n}\chi) - (R^{B,\mathrm{vol}}_{K,n}, C^{\mathrm{tf},-}_{K,n}\chi)_{Q_{K,n}} \qquad \forall \chi \in Q^{q_t}_K.\end{aligned} \tag{38}$$

No spatial-facet subtraction appears because $b_K = 0$ on ∂K. For the usual one-sided dG formulation, the recovered temporal residual is assigned in full to the following slab I_n.

6.3 Splitting the Recovered Residual with Numerical Dual Weights

Bubble recovery supplies computable residual densities, but it does not determine whether their contribution should trigger spatial (h) or temporal (k) refinement. That directional separation is provided by the dual weights.

Theoretical error decomposition. For a linear goal functional $\mathcal{J}(\cdot)$, the global error can be conceptually split by introducing a time-half-discrete intermediate solution U_k (discrete in time, continuous in space):

$$\mathcal{J}(u) - \mathcal{J}(U_{kh}) = \underbrace{\mathcal{J}(u) - \mathcal{J}(U_k)}_{e_k \text{ (temporal error)}} + \underbrace{\mathcal{J}(U_k) - \mathcal{J}(U_{kh})}_{e_h \text{ (spatial error)}}. \tag{39}$$

In computation, exact dual solutions are replaced by discrete adjoints evaluated on enriched spaces:

- $Z_{00} := Z_{kh}$: standard fully discrete adjoint on the current space–time mesh;
- $Z_{01} := Z_{hk^+}$: adjoint with time-enriched discretisation (k^+);
- $Z_{10} := Z_{h^+k}$: adjoint with space-enriched discretisation (h^+);
- $Z_{11} := Z_{h^+k^+}$: fully enriched space–time adjoint.

One can directly follow the time-first splitting corresponding to equation (39) by introducing only the time-enriched adjoint Z_{01}:

$$w_h := Z_{11} - Z_{01}, \tag{40}$$
$$w_k := Z_{01} - Z_{00}. \tag{41}$$

This option requires solving three adjoint problems (Z_{00}, Z_{01}, Z_{11}). Analogously, a space-first split using Z_{10} defines $w_h := Z_{10} - Z_{00}$ and $w_k := Z_{11} - Z_{10}$.

6.4 Local Directional Indicators

Rather than artificially decomposing the residual function itself into purely spatial or temporal components, directional refinement is driven by testing the entire recovered residual vector—volumetric, spatial-facet, and temporal-interface—against the respective dual weights w_h and w_k.

For each refinement direction $\alpha \in \{h, k\}$, define the signed entity-level error contributions:

$$\eta_{K,n}^{\mathrm{vol},\alpha} := (R_{K,n}^{B,\mathrm{vol}}, w_\alpha)_{Q_{K,n}}, \tag{42}$$
$$\eta_{F,n}^{\mathrm{sf},\alpha} := (R_{F,n}^{B,\mathrm{sf}}, \{w_\alpha\})_{F \times I_n}, \tag{43}$$
$$\eta_{K,n-1}^{\mathrm{tf},\alpha} := (R_{K,n-1}^{B,\mathrm{tf}}, w_\alpha(t_{n-1}^+))_K, \tag{44}$$

where $\{w_\alpha\}$ denotes the appropriate trace or average defined by the spatial scheme, and boundary facet terms are integrated with their boundary trace.

To localise these contributions onto individual space–time slabs $Q_{K,n} = K \times I_n$, ownership of interface residuals is assigned according to their topological continuity:

- **Spatial Facets (Half-Splitting):** An interior spatial facet $F \in \partial K^+ \cap \partial K^-$ is shared by two adjacent spatial elements. Its contribution is split equally ($\frac{1}{2}$) between the two neighbouring cell ledgers.

- **Temporal Interfaces (One-Sided Assignment):** Due to the directional causality of time in dG formulations, the interface jump $R^{B,\mathrm{tf}}_{K,n-1}$ at t^{+}_{n-1} acts as the initial condition mismatch for the current slab I_n. It is therefore assigned in full (100%) to I_n without temporal half-splitting.

Combining these contributions yields the net local indicator ledger for direction α on cell K in slab I_n:

$$\boxed{\eta^{\alpha}_{K,n} := \eta^{\mathrm{vol},\alpha}_{K,n} + \frac{1}{2}\sum_{F\subset\partial K\cap\Omega} \eta^{\mathrm{sf},\alpha}_{F,n} + \sum_{F\subset\partial K\cap\partial\Omega} \eta^{\mathrm{bdry},\alpha}_{F,n} + \eta^{\mathrm{tf},\alpha}_{K,n-1}.} \tag{45}$$

Finally, the adaptive algorithm extracts two distinct indicator families to guide spatial mesh adaptation and time-step control separately:

$$\boxed{\eta^{h,\mathrm{loc}}_{K,n} := \eta^{h}_{K,n}, \quad K \in \mathcal{T}^n_h,\ n = 1,\dots,N,} \tag{46}$$

$$\boxed{\eta^{k,\mathrm{loc}}_{n} := \sum_{K\in\mathcal{T}^n_h} \eta^{k}_{K,n}, \quad n = 1,\dots,N.} \tag{47}$$

Consequently, $\eta^{h,\mathrm{loc}}_{K,n}$ identifies specific spatial elements K on mesh $\mathcal{T}^n_h$ that require h-refinement, whereas $\eta^{k,\mathrm{loc}}_{n}$ aggregates spatial contributions across the domain to determine if the time step k_n of interval I_n should be bisected or enlarged. Residual geometry and refinement direction remain strictly decoupled throughout the localisation pipeline.

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